Jonathan Haberer

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Automation Engineer | AEC Systems

Automation engineer with six years inside a production AEC firm. I embed with the team doing the work, model how the workflow actually operates, then automate it end to end. Promoted from junior drafter to Disaster Recovery Engineering Manager / Head of R&D in four years. The systems I built scaled the firm from ~10 projects a week to 100+ per day.

WORK EXPERIENCE

Cobalt Engineering and Inspections — **August 2020 – Present**

Disaster Recovery Engineering Manager / Head of R&D — *Sep 2024 – Present*

Disaster Recovery Engineering Manager — *Aug 2023 – Sep 2024*

- Own engineering delivery for federally funded disaster recovery programs in eight states, including Renew NC ($1.4B), LeeCares ($1.1B), Better Future ($585.7M), Restore Louisiana, ReCoverCA, and Texas GLO. Approve all engineering and drafting output and serve as the program-facing lead with state recovery offices, including governor's office program validation.
- Remain the working engineer on complex projects: personally design and QC structural packages for custom and restricted program homes (foundations, framing, and wind design in coastal high-wind and flood jurisdictions), with final approval on every plan the department ships.
- Lead a distributed team of 7 engineers and drafters plus 2 managers. Recent intake hit 300+ projects in two weeks; the automation can produce 100+ program homes per day.
- Nine days from first shadowing the survey/civil team to their pipeline running automated: a C#/.NET plugin inside Civil 3D with a five-layer OCR pipeline (Windows OCR, PaddleOCR, Tesseract, custom assembly engine) for deed and plat extraction from documents dating to the early 1900s. Parcel turnaround dropped from ~2 weeks to ~1, the standing backlog cleared, and every extraction now ships with an auditable RPLS review pack.

Lead Structural Designer / Senior CAD Designer / PM — *Mar 2022 – Aug 2023*

- Hands-on structural design lead for the portfolio's complex work: Tyndall AFB recovery structures (hangars, communications facilities, bunker, vault) and Schlitterbahn Galveston, where I led design of the support structures for two tower systems and six slides. Oversaw final construction documents and code compliance across jurisdictions.
- Designed and shipped the client intake portal (React front end → Azure microservice API gateway → QuickBase), replacing email-and-PDF intake. Now on v4, serving 220+ client organizations and 16 internal staff, absorbing peak intake of 10 projects every 5 minutes.

Structural Designer / Engineering Technician / PM — *Aug 2021 – Mar 2022*

- Designed residential and commercial structures, including complex foundation and framing systems, with structural evaluations in RISA 3D, ClearCalcs, and custom Excel tooling. Performed on-site inspections and coordinated directly with county officials to expedite disaster rebuilds.
- Sole developer of "Program Squared," a six-month build merging calculations with CAD: AutoLISP, AutoCAD scripting, PowerShell, and batch under a Java runtime that generates site-specific architectural and structural plan sets in 3-4 minutes versus ~2 hours manual. Pulls ASCE hazard data per address via GIS overlay so member and sheathing selection is correct for every site, engineering rigor the manual process skipped.
- Took custom-plan throughput from ~2/month to ~20/month with no added headcount.

Structural Drafter / PM | Junior Drafter / Structural Technician — *Aug 2020 – Aug 2021*

- Drafted 100+ custom homes and 400+ Florida disaster recovery homes. Field work included LiDAR scanning, ground-penetrating radar, and on-site drafting, plus compliance work for historical renovations and protected environments.

INDEPENDENT WORK

Firmark, Founder / Sole Engineer — *2023 – Present*

- Solo-built a browser-based CAD engine and structural calculation engine (live demos at tryfirmark.com), plus a closed automation engine that turns an architectural plan into calculations, a bill of materials, and PE-stampable plan sets in minutes, with human approval gates at every stage. Sample automated outputs are viewable inside the drafting demo.

Region Proposal Networks for Utility Asset Detection (technical write-up)

- Faster R-CNN / RPN analysis for pole-component detection: anchor strategy for extreme aspect ratios, k-means anchor optimization, FPN tradeoffs, edge deployment design, and a hands-on ResNet50 feature-map and region-proposal experiment. (read the write-up)

SKILLS

- **Languages:** C#, Python, Java, JavaScript, AutoLISP, PowerShell
- **Platforms & APIs:** .NET, Civil 3D API, Revit API, Dynamo, Forge/APS, BIM 360, Azure, QuickBase, Google Maps API
- **ML & CV:** PyTorch, torchvision, Faster R-CNN/RPN, ResNet50, OpenCV, Tesseract, PaddleOCR
- **Engineering & CAD:** AutoCAD (2D/3D/Civil), Civil 3D, Revit, RISA 3D, ClearCalcs, Bluebeam, Navisworks, SketchUp, SDS2; LiDAR, GPR, structural evaluation

EDUCATION

- San Jacinto College: returned for engineering coursework toward a second A.S., completed all but one semester. Left to pursue full-time engineering and automation work.
- A.S., General Studies, earned concurrently with my high school diploma at Clear Horizons Early College High School (on the San Jacinto College South campus); graduated early, December 2017.